JOHN WICK UNIVERSE DATASET
EDA AND REGULARIZATION REPORT

Dataset Shape: (96176, 12)

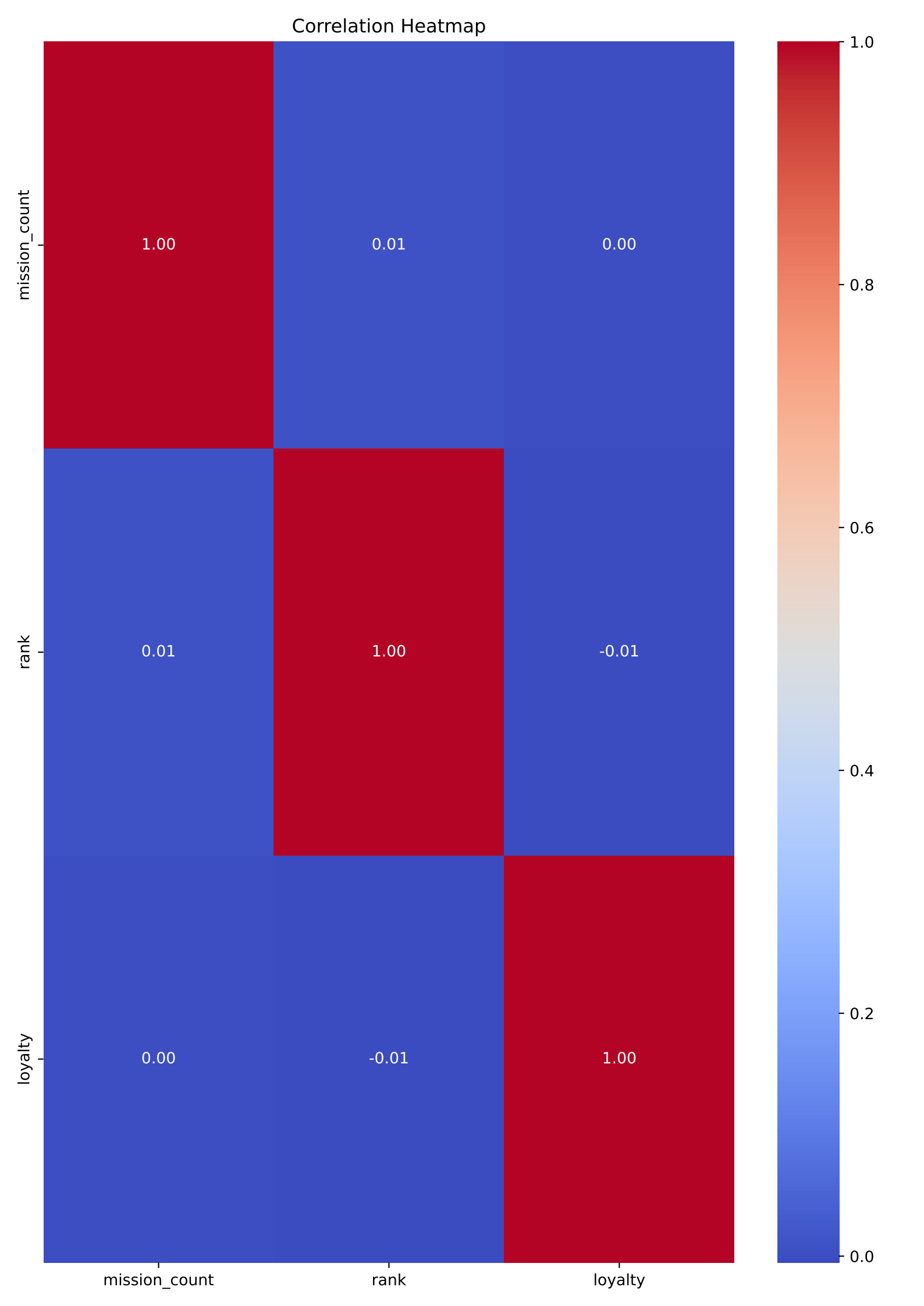
Target Variable: loyalty

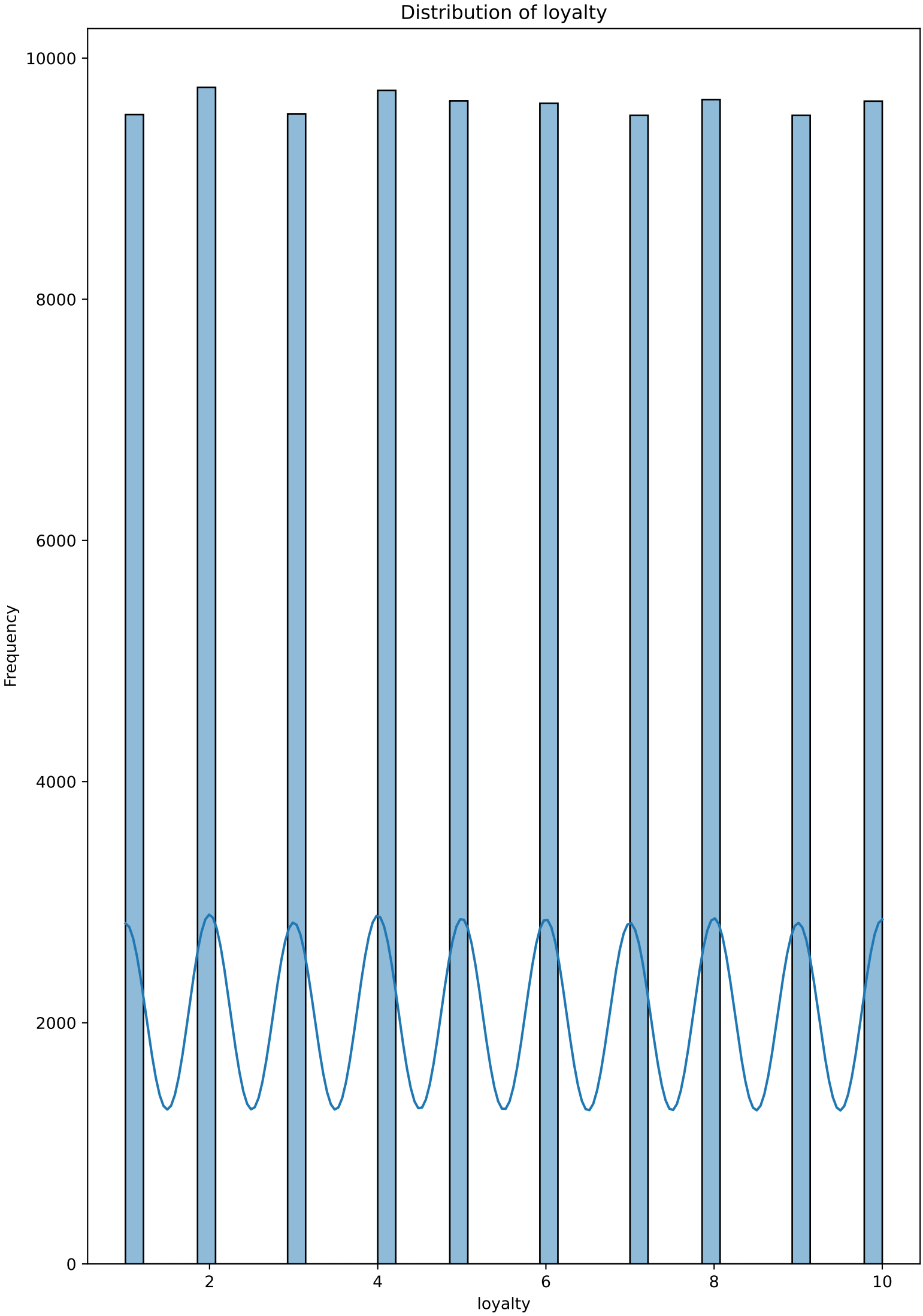
Numerical Features: 2
Categorical Features: 9

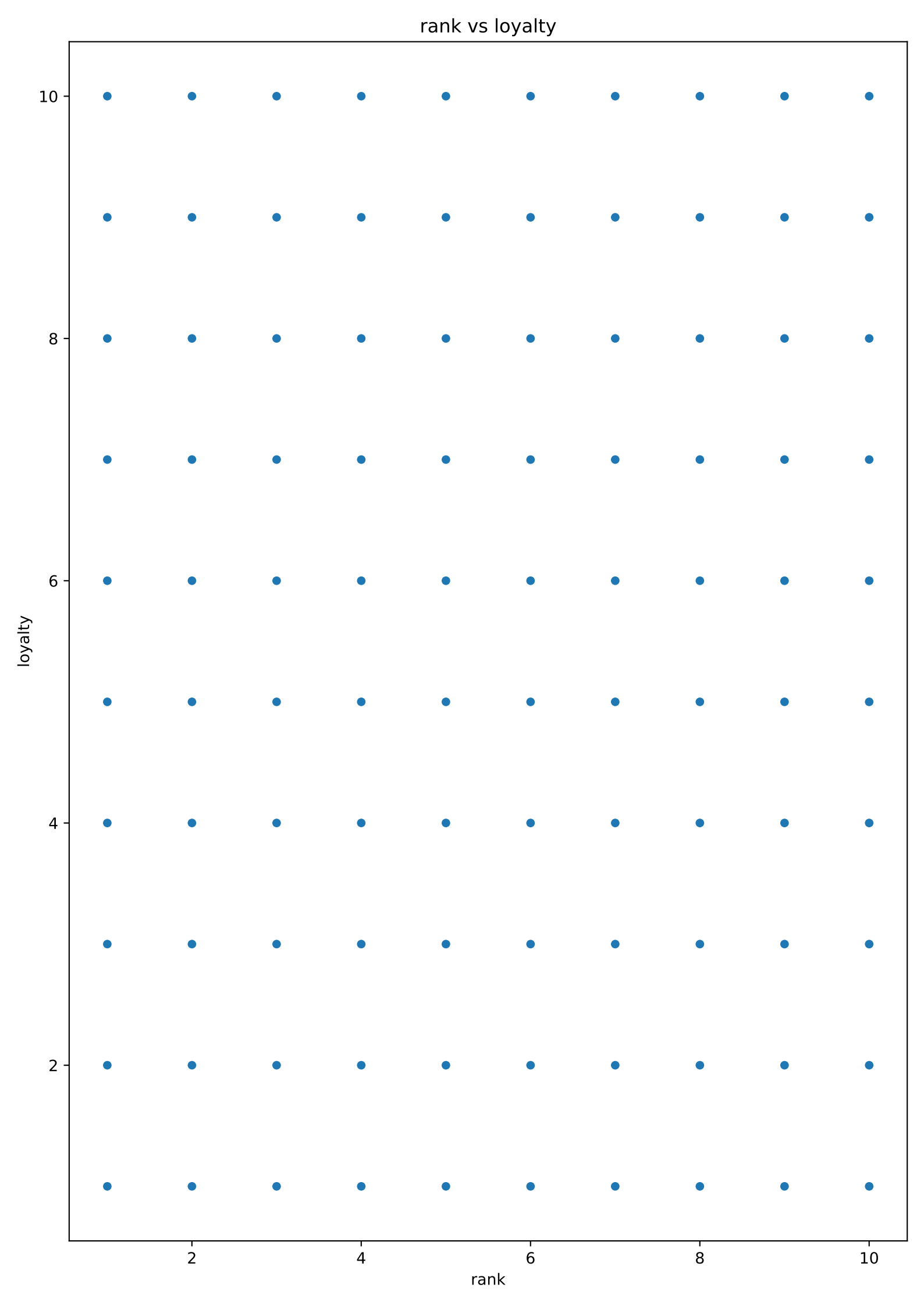
Missing Values: 0
Duplicate Rows: 4

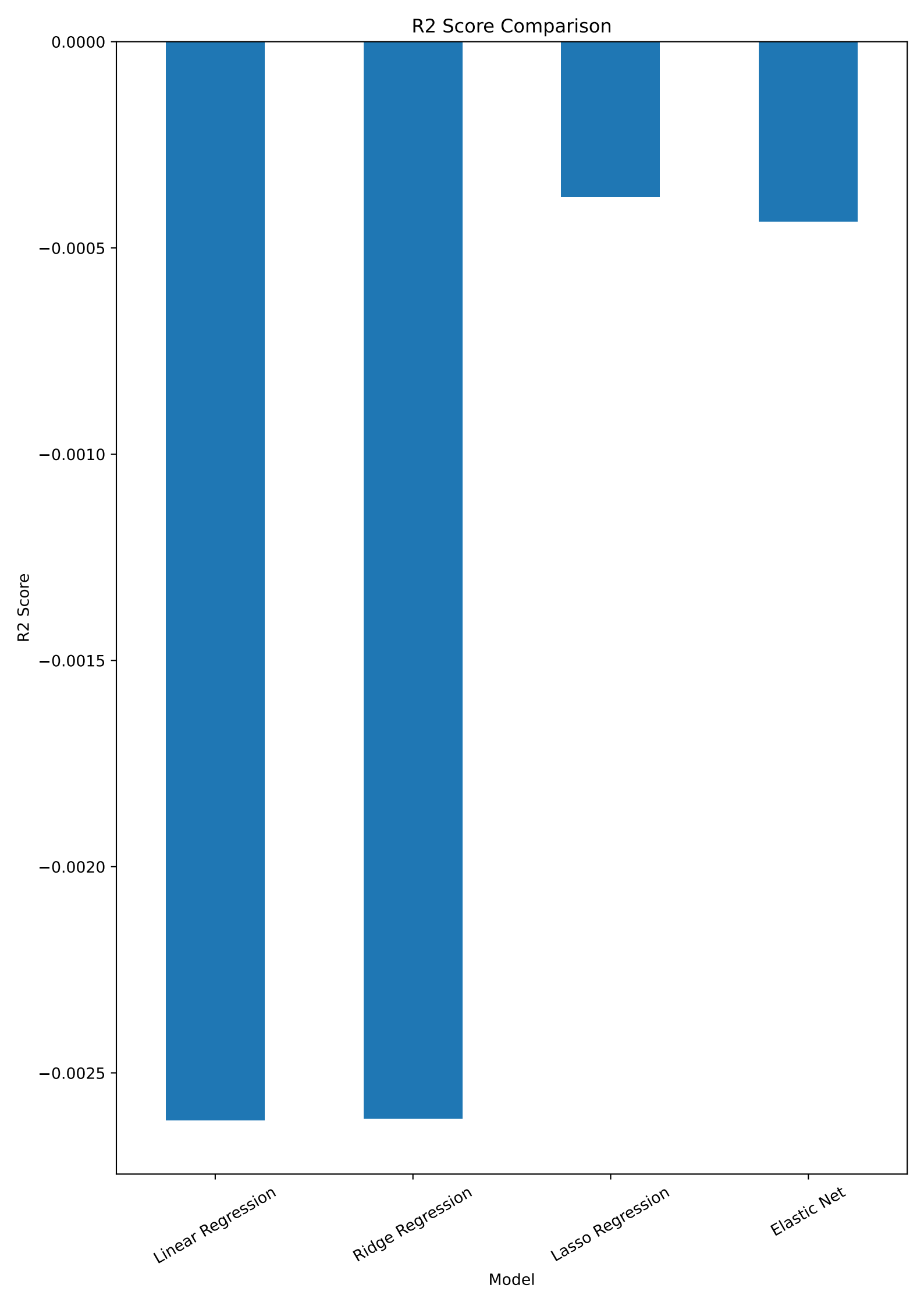
MODEL PERFORMANCE

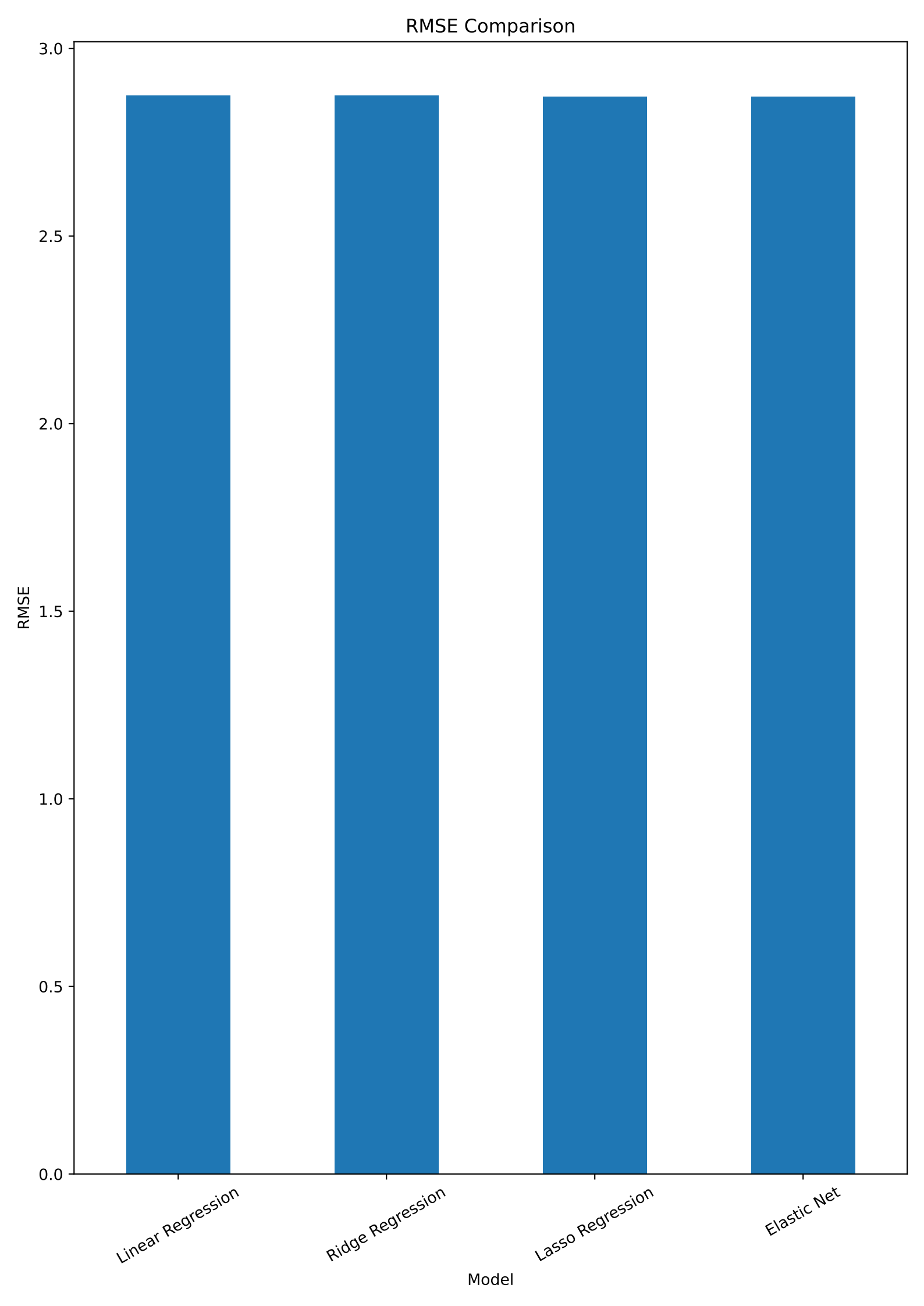
	MAE	MSE	RMSE	R2 Score
Linear Regression	2.4983	8.2607	2.8741	-0.0026
Ridge Regression	2.4983	8.2607	2.8741	-0.0026
Lasso Regression	2.4983	8.2423	2.8709	-0.0004
Elastic Net	2.4983	8.2427	2.8710	-0.0004

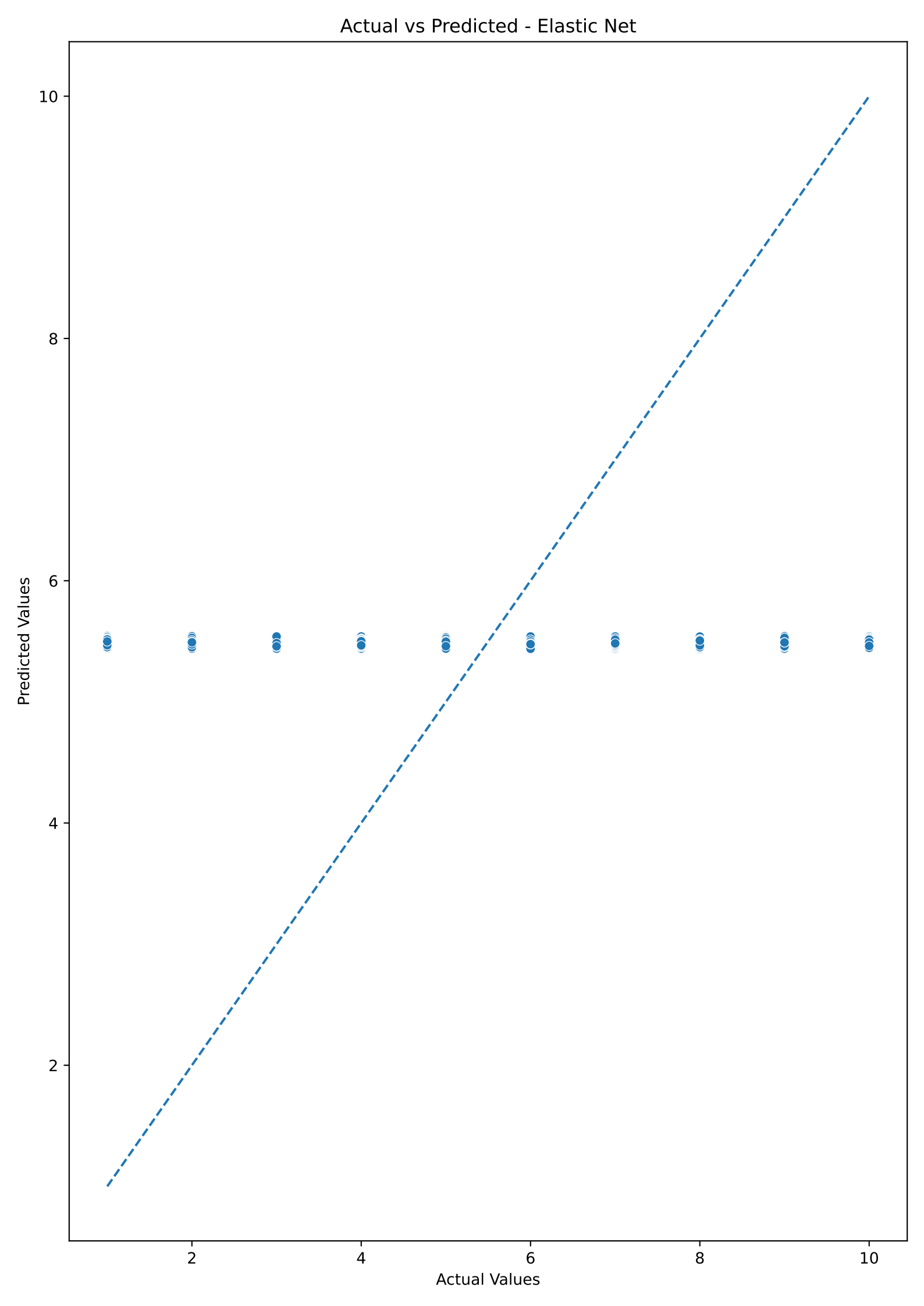

Correlation Heatmap
mission_count
rank
loyalty
1.00
0.01
0.00
0.01
1.00
-0.01
0.00
-0.01
1.00
mission_count
rank
loyalty
1.0
0.8
0.6
0.4
0.2
0.0


Distribution of loyalty
Frequency
loyalty
0
2000
4000
6000
8000
10000
2
4
6
8
10


rank vs loyalty
loyalty
rank
10
8
6
4
2
2
4
6
8
10


R2 Score Comparison
0.0000
−0.0005
−0.0010
−0.0015
−0.0020
−0.0025
R2 Score
Linear Regression
Ridge Regression
Lasso Regression
Elastic Net
Model


RMSE Comparison
3.0
2.5
2.0
1.5
1.0
0.5
0.0
RMSE
Linear Regression
Ridge Regression
Lasso Regression
Elastic Net
Model


Actual vs Predicted - Elastic Net
Predicted Values
Actual Values
2
4
6
8
10

FINAL OBSERVATIONS

Best Model: Lasso Regression

Best R2 Score: -0.0004

Best RMSE: 2.8709

Lasso Zero Coefficients: 158

Elastic Net Zero Coefficients: 153

Ridge Regression applies L2 regularization.
Lasso Regression applies L1 regularization.
Elastic Net combines L1 and L2 regularization.